

Kwangho Baek, Ph.D.

Postdoctoral Associate in Transportation Engineering

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RESEARCH PROFILE

My research advances multimodal transportation planning through behavioral models and system optimization. I develop frameworks that integrate artificial intelligence (AI) and machine learning into established approaches, including four-step and activity-based models, while preserving their methodological strengths and compatibility with planning practice. These frameworks harness expanding transportation data and computational capabilities to support adaptive, equitable planning and evidence-based policy decisions.

Research Areas Travel Behavior and Demand Modeling; Multimodal Transportation Planning; Transit Operations and Data Analytics; Rural Mobility; Mobility-as-a-Service (MaaS); Transportation Equity; Shared Mobility.

Methods/Tools Discrete Choice Modeling; Machine Learning; Reinforcement Learning; Generative AI; Spatial and Network Analysis; Python; R; PyTorch; scikit-learn; ArcGIS/QGIS; SQL; EMME4; TransCAD; Synchro.

EDUCATION

Ph.D. and Master of Science in Civil (Transportation) Engineering

- Dissertation: *Toward Adaptive Transit Planning with Artificial Intelligence*.

- Advisor: Dr. Alireza Khani. GPA: 3.905/4.000.

University of Minnesota, Twin Cities

Sep 2021–Feb 2026

Master of Science in Urban Planning and Engineering

- Thesis: *The Impact of the Bus Crowding Information System on Passenger Bus Choice*.

- Advisor: Dr. Jin-Hyuk Chung. GPA: 4.21/4.30.

Yonsei University, Seoul

Mar 2018–Feb 2020

Bachelor of Science in Urban Planning and Engineering

- Capstone: *A Study on the Urban Development in North Korea After Reunification: Focused on Micro-districts in Pyongyang*. Co-authors: Hyunmin Daniel Zoh, Subin Park, and Jihyeon Ock.

- GPA: 3.76/4.30, *magna cum laude*.

Yonsei University, Seoul

Mar 2012–Feb 2018

PEER-REVIEWED PUBLICATIONS

Underlined name: Kwangho Baek; * denotes corresponding author.

Published or In Press

- **Navigating Machine Learning for Travel Behavior Analysis: A Comprehensive Guide to Inferring Trip Purposes Using Transit Survey and Automatic Fare Collection Data Fusion**
 - Kwangho Baek and Alireza Khani*
 - *Transportation Research Part C: Emerging Technologies* (2026). Vol 190. 105726.
- **Evaluating the Effects of Mobility-as-a-Service Platform in Increasing Equity and Transit Ridership in Rural Areas**
 - Kwangho Baek and Alireza Khani*
 - *Transportation Research Record* (2026). Vol Online First.
Also presented at the *Transportation Research Board 104th Annual Meeting*. Washington, DC. Jan 6, 2025 (Lectern)
- **Charger Deployment and Fleet Sizing for Heterogeneous Fleets Under Budget-Constrained Multi-Stage Electrification**
 - Nastaran Tork, Behnam Emami, Kwangho Baek, Alireza Khani*, Sushmita Bhandari, and Alyssa Ryan
 - *Transport Policy* (2026). Vol 187, 104285.
Also presented at the *Transportation Research Board 105th Annual Meeting*. Washington, DC. Jan 12, 2026 (Lectern)
- **Efficient Urban Transit Systems: Bridging Freight Logistics and Passenger Transport**
 - Nastaran Tork, Kwangho Baek, and Alireza Khani*
 - *Transportation Research Record* (2025). Vol 2679(11), 782–804.
Also presented at the *Transportation Research Board 104th Annual Meeting*. Washington, DC. Jan 6, 2025 (Lectern)
- **Electric Scooter Sharing: How Do People Value It as a Last-Mile Transportation Mode?**
 - Kwangho Baek, Hyukseong Lee, Jin-Hyuk Chung, and Jinhee Kim*
 - *Transportation Research Part D: Transport and Environment* (2021). Vol 90, 102641.
- **Factors Affecting Heterogeneity in Willingness to Use E-Scooter Sharing Services**
 - Hyukseong Lee, Kwangho Baek, Jin-Hyuk Chung, and Jinhee Kim*
 - *Transportation Research Part D: Transport and Environment* (2021). Vol 92, 102751.

Under Revision or Review

- **Unlocking Vehicle-Miles-Traveled Potential: Behavioral Segmentation and Trip-Context Effects from a Machine Learning-Augmented Mode Choice Model**
 - Kwangho Baek and Alireza Khani*
 - Under Revision, *Transportation Research Record*.
Also accepted for presentation at the *Transportation Research Board 106th Annual Meeting*. Washington, DC. Jan 2027
- **Supporting High-Speed Rail with a Hub-and-Shuttle Approach: Transforming Long-Haul Conventional Rails into Short and Frequent Feeders**
 - Ju-Yeon Lee, Kyungtaek Kim, and Kwangho Baek*
 - Under Revision, *Transportation Research Procedia*.
Also presented at the *World Conference on Transport Research (WCTR)*. Toulouse, France. Jul 10, 2026 (Poster)
- **A Discrete Choice Model with Segmentation and Embedding-Augmented Learning for Travel Behavior Analysis**
 - Kwangho Baek and Alireza Khani*
 - Under Review, *Transportation*.
Also presented at the *Conference on Advanced Systems in Public Transport (CASPT)*. Kyoto, Japan. Jul 1, 2025 (Poster)

CONFERENCE PRESENTATIONS (PEER-REVIEWED)

- **Learning-Based Intelligent and Adaptive Dispatching for MaaS-Enabled Intermodal Rural Transit**
 - Kwangho Baek and Alireza Khani*
 - (Accepted) *Transportation Research Board 106th Annual Meeting*. Washington, DC. Jan 2027.
- **Evaluating Incremental Truck Weight Limit Policy at the Statewide Network Scale: A Minnesota Case Study**
 - Nastaran Tork, Alireza Khani*, Kwangho Baek, Mihai Marasteanu
 - (Accepted) *Transportation Research Board 106th Annual Meeting*. Washington, DC. Jan 2027.
- **Quantifying Pavement Damage Costs Associated with Increased Axle Loads Using Life-Cycle Cost Analysis**
 - Zifeng Zhao*, Kaidi Liang, Kwangho Baek, Mihai Marasteanu, and Alireza Khani
 - (Accepted) *Transportation Research Board 106th Annual Meeting*. Washington, DC. Jan 2027.
- **Enhancing Rural Transit Efficiency through MaaS Integration of Demand-Responsive Transit and Fixed Routes**
 - Kwangho Baek and Alireza Khani*
 - *ASCE International Conference on Transportation and Development (ICTD)*. Detroit, MI. Jul 1, 2026. (Lectern)
- **Dynamic Evolution of MaaS Subscriptions: A Game-Theoretic Day-to-Day Framework for Multimodal Transport**
 - Jiale Li, Alireza Khani*, and Kwangho Baek
 - *ASCE International Conference on Transportation and Development (ICTD)*. Detroit, MI. Jun 30, 2026. (Poster)
- **Barriers to Local Government Electric Vehicle Fleet Transitions: Insights from Minnesota**
 - Sushmita Bhandari, Juliana Panhorst, Alyssa Ryan*, Nastaran Tork, Kwangho Baek, and Alireza Khani
 - *Transportation Research Board 105th Annual Meeting*. Washington, DC. Jan 12, 2026. (Lectern)
- **I Want More Than a Bus: Using AI-Integrated Discrete Choice Model to Identify Likely Users of Premium Transit**
 - Kwangho Baek and Alireza Khani*
 - *2025 Modeling Mobility (MoMo) Conference*. Minneapolis, MN. Sep 16, 2025. (Lectern)
- **The Changes in Transit Route Choice Behavior Between Pre- and Post-Pandemic**
 - Sahas Sok, Kwangho Baek, and Alireza Khani*
 - *Transportation Research Board 104th Annual Meeting*. Washington, DC. Jan 8, 2025. (Poster)
- **Integrating Neural Networks and Latent Class Choice Model to Analyze Transit Path Choice: Understanding Likely Users of Branded or Premium Transit Routes**
 - Kwangho Baek and Alireza Khani*
 - *Transportation Research Board 104th Annual Meeting*. Washington, DC. Jan 7, 2025. (Poster)
- **Changes in Pre- and Post-Pandemic Transit Commute Trips: Findings from Integrating Smart Card Data and Passenger Surveys with Machine Learning Models**
 - Kwangho Baek and Alireza Khani*
 - *Transit Data 2024*. London, United Kingdom. Jul 3, 2024. (Lectern)
- **A DRT-Bus Intermodal Rural Transit Operation Powered by MaaS Applying Differentiated Dispatch Priorities**
 - Kwangho Baek and Alireza Khani*
 - *Transportation Research Board 103rd Annual Meeting*. Washington, DC. Jan 9, 2024. (Lectern)
- **How Do People Value Reliable Modes in Their Transit Path Choices?**
 - Kwangho Baek and Alireza Khani*
 - *International Symposium on Transport Network Resilience (INSTR)*. Hong Kong. Dec 13, 2023. (Lectern)
- **How Has Real-Time Crowding Information Affected Passengers' Bus Choice?**
 - Kwangho Baek and Jin-Hyuk Chung*
 - *Transportation Research Board 101st Annual Meeting*. Washington, DC. Jan 12, 2022. (Lectern)

- **A Practical Method to Adjust Bus Routes Based on Transfer Penalties Using Trip-Chain Data and SP Survey**
- Younghun Bahk, Kwangho Baek, and Jin-Hyuk Chung*
- *Transportation Research Board 100th Annual Meeting*. Washington, DC (virtual). Jan 28, 2021. (Online)

INVITED TALKS

- **Longitudinal Analysis of Transit Rider Behavior Using AFC and Survey Data Fusion via Machine Learning: COVID-19-Driven Commute Pattern Changes**
- Kwangho Baek and Alireza Khani
- *Transportation Research Board 105th Annual Meeting*. Washington, DC. Jan 12, 2026.
- **Impacts of Expanding Transit Data Access for Rural Transit in Minnesota**
- Elliott McFadden, Kwangho Baek, Hannah DeBruin, and Alireza Khani
- *2024 CTS Transportation Research Conference*. Minneapolis, MN. Nov 7, 2024.

TECHNICAL REPORTS

- **Transitioning to EV Fleets: Best Practices and a Decision Tool**
- Nastaran Tork, Kwangho Baek, Alireza Khani, Sushmita Bhandari, and Alyssa Ryan
- *Minnesota Department of Transportation* (2026). Report No. MN 2026-15.
- **Transportation Needs and Current Services in the 5-County East Central Region**
- Kwangho Baek, Theodore Jessup, Frank Douma, Alireza Khani, and Michael Darger
- *Center for Transportation Studies, University of Minnesota* (2025). Report No. ESMC 26-01.
- **Transfer Behavior and Off-Peak Commutes**
- Kwangho Baek and Alireza Khani
- *Center for Transportation Studies, University of Minnesota* (2024). Report No. CTS 24-06.
- **MnDOT's Mobility-as-a-Service Platform: Assessing User Behavior and Measuring System Benefits**
- Kwangho Baek, Hannah DeBruin, Alireza Khani, and Elliott McFadden
- *Center for Transportation Studies, University of Minnesota* (2024). Report No. CTS 24-01.

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Postdoctoral Associate, Transit Lab

University of Minnesota, Twin Cities

Mar 2026–Present

- Manage research projects and instruct graduate students/undergraduate interns.
- Develop research proposals and pursue external funding opportunities.
- Develop integrated multimodal routing, choice modeling, and generative AI frameworks to advance activity-based modeling and inform transportation planning and policy. Funded by *Minnesota DOT*.
- Investigate the effects of transportation infrastructure on economic development through benefit-cost analysis, accessibility modeling, and simulation. Funded by *Minnesota DOT*.

Graduate Assistant and Lab Manager, Transit Lab

University of Minnesota, Twin Cities

Aug 2021–Feb 2026

- Collaborated with graduate students and undergraduate interns on research projects and managed laboratory equipment, servers, databases, and software tools.
- Conducted a regional transit needs assessment using spatial gap analysis in East Central Minnesota. Funded by the *Empowering Small Minnesota Communities (ESMC) Program*.
- Evaluated the effects of Mobility-as-a-Service deployment on rural transit in Minnesota. Funded by *Minnesota DOT* via *Federal Transit Administration (FTA)* grant.
- Investigated the effects of enhanced transit services on off-peak commuting by developing transit pathfinding algorithms and integrating automatic fare collection data with passenger surveys through machine learning. Funded by the *Metropolitan Council (Minnesota)*.

Researcher, Department of Rail Transport

Korea Transport Institute (KOTI)

Dec 2019–Jul 2021

- Conducted demand and feasibility analysis for the Fourth National Railway Plans of Korea (2021–2030) using EMME4, funded by *Korea Ministry of Land, Transportation, and Infrastructure (MOLIT)*.
- Proposed operation-speed-based resource reallocation for Korean National Rail, optimizing a mixed low- and high-speed rail network under vehicle and budget constraints, funded by Korea MOLIT.

Graduate Assistant and Lab Manager, Transportation Planning Lab

Yonsei University, Seoul

Feb 2018–Dec 2019

- Managed laboratory equipment and personnel and trained incoming graduate students.
- Developed a structural equation model to examine factors associated with transit user satisfaction nationwide. Funded by *Korea Transportation Safety Authority (TS)*.
- Conducted network analyses to identify an optimal national firefighters' hospital location, considering weighted multimodal access time, revenue, and healthcare equity. Funded by the *National Fire Agency of Korea*.

Research Assistant

Barun ICT Research Center

- Engineered an ArcGIS geoprocessing algorithm for identifying geospatial clusters of large-scale social media location data. Aug 2017–Feb 2018

Undergraduate Intern, Environmental Spatial Informatics Lab

Yonsei University, Seoul

- Developed ArcGIS instructional materials for a 3-credit 3000-level undergraduate course.
- Supported NLP-based text mining to extract disaster events from historical news data.

Dec 2016–Mar 2017

Terrain and GIS Analyst as a Military Engineer (Sergeant)

Republic of Korea Army

- Managed GIS databases and created on-demand tactical maps for an infantry division.
- Developed an ArcGIS algorithm to detect eligible battlefield water points.

Apr 2014–Jan 2016

TEACHING EXPERIENCE

Lab Instructor (CEGE 3201: Transportation Engineering)

University of Minnesota, Twin Cities

- Taught weekly one-hour laboratory sessions in English using R and Synchro to teach traffic engineering and four-step travel demand modeling.
- Mean student teaching evaluation across two semesters: 5.48/6.00.
- Student feedback (Fall 2025): *"He breaks down concepts clearly and takes time to work through examples step by step. His patience with students who have no coding background really stands out."*
- Student feedback (Fall 2022): *"I liked that he would guide us through how to use commands in R. He explained everything he was doing and made it easy to follow along. Great instructor!"*

Fall 2022; Fall 2025

Lab Instructor (CRP 3000: Transportation System Analysis)

Yonsei University, Seoul

- Taught weekly one-hour laboratory sessions in Korean using R and TransCAD to teach and demonstrate four-step travel demand modeling with on-Campus pedestrian movement data.

Spring 2019

Teaching Assistant (Grading and Mentoring)

Yonsei University, Seoul

- CRP 2320: Transportation Planning (undergraduate class)
- CRP 6561: Transportation Planning (graduate class)
- CRP 3660: Site Transportation Planning

Spring/Fall 2018

PROFESSIONAL SERVICE, MEMBERSHIPS, AND HONORS

Journal and Conference Peer Review

- Transportation Research Part D: Transport and Environment:** Reviewed 2 journal manuscripts.
- Transportation:** Reviewed 1 journal manuscript.
- Transportation Letters:** Reviewed 1 journal manuscript.
- Transportation Research Today:** Reviewed 1 journal manuscript.
- Transportation Research Board (TRB) Annual Meeting:** Reviewed 12 conference papers (7500 words each).
- International Symposium on Transport Network Resilience (INSTR):** Reviewed 2 extended abstracts.

Academic Service and Professional Memberships

Jun 2026–Dec 2029	World Conference on Transport Research Society (WCTRS)	Member
Feb 2026–Dec 2027	American Society of Civil Engineers (ASCE)	Member
Jan 2022–Jan 2027	Korean-American Scientists and Engineers Association (KSEA)	Student Member
Sep 2021–Feb 2026	Minnesota Korean Graduate Students Association (MKGSA)	Department Representative
Sep 2024–Aug 2025	UMN CEGE Graduate Student Council	Academic Chair/Transportation Representative
Mar 2018–Dec 2019	Korean Society of Transportation (KST)	Student Member

Honors and Awards

2026	Robert C. Johns Research Partnership Award. (team award) Center for Transportation Studies.
2025	Matthew J. Huber Award for Excellence in Transportation Research. Center for Transportation Studies.
2025; 2022	Best Presentation Awards. CEGE 8200 Transportation Seminar, University of Minnesota.
2024	Excellent Presentation Award. Korea Transportation Association in America.
2019	Brain Korea 21c Fellowship. Korea Ministry of Education.
2018	President's Award. Korean Society of Transportation.